Twice-played prisoners' dilemma

Suppose that the prisoners' dilemma in Figure 1 is played twice. At date 1, the two players simultaneously choose

		Player 2 C	Player 2 D
Player 1	C	$2,2$	$-1,3$
	D	$3,-1$	$0,0$

Figure 1: Prisoners' dilemma stage game

C or D. After observing the date-1 choice, the players again simultaneously choose C or D. The game tree is in Figure 2. A *strategy* for a player is this game must specifiy five things, since each player has five information sets:

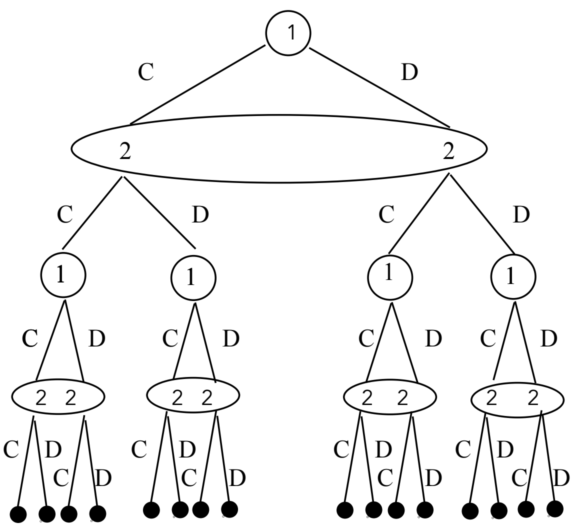

Figure 2: Extensive form for twice-played prisoners' dilemma. Assume that the utility of *outcome abcd* is the sum of the utility numbers in the stage game for *ab*—the date-1 choice—and *cd*—the date-2 choice—, where each letter is either C or D.

what the player does at date 1; and what the player intends to do after each of the *four* possible date-1 histories. Since each player has two possible actions at each of her information sets, each player has *thirty-two* strategies in this game: the strategic-form matrix of this game would have thirty-two rows and thirty-two columns.